Price elasticity and the cinema pricing decision

Teacher guidance and answers | Original TBS classroom practice

Use this lesson after students have met the PED formula, the meaning of elastic and inelastic demand and the factors that influence PED. Students calculate PED for two products, link it to revenue, and then test the plan against contribution and the limits of the evidence. The tasks are planned to take about 55 minutes and have not been trialled in a classroom. Allow about 65 minutes in the timetable for handing out, whole-class feedback and the plenary. Suitable for classes studying Edexcel 9BS0 Theme 1, in Year 12 or as Year 13 revision.

Curriculum fit

Pearson Edexcel A level Business 9BS0, Specification Issue 1 (September 2014), Theme 1 (Marketing and people), section 1.2.4 Price elasticity of demand: (a) calculation of PED, (b) interpretation of numerical values, (c) the factors influencing PED, (d) the significance of PED to businesses in terms of pricing and (e) calculation and interpretation of the relationship between PED and total revenue. Task 3 adds contribution (Theme 2, 2.2.3(a), defined in the specification per unit; the lesson applies it to weekly totals) as an extension of the revenue idea. Checked against the archived Issue 1 specification; the live Pearson course page was opened on 8 October 2026 and showed no withdrawal notice in the fetched text, so please confirm the specification for your cohort with your exams officer. This is a TBS lesson, not a Pearson resource; it is not endorsed by Pearson and uses no Pearson exam questions or mark scheme.

Sign convention: the lesson reports PED with its negative sign, as for almost all goods, and compares the size (magnitude) with 1. The specification does not prescribe a sign convention; accept an answer that omits the minus sign if the student says it is the size of PED. All business details and numbers are fictional. The answers are indicative teaching guidance only. There is no official mark tariff, level descriptor or AO allocation for these tasks and no grade claim is made.

Lesson plan (about 55 minutes of tasks, estimated)

Stage	Minutes	What happens
Starter: elastic or inelastic?	5	Classify four PED values.
Task 1: tickets	10	Percentage changes, PED, revenue.
Task 2: snack combo	8	PED, revenue and the link to pricing.
Task 3: revenue and contribution	12	Contribution table and comparison.
Task 4: why different?	8	Factors influencing PED.
Task 5: decide	8	Evaluate the manager's claim.
Exit ticket	4	A price rise test.
Total	55	

45-minute version: starter 4, Tasks 1 and 2 at the same time 8 (give Task 1 to half the class and Task 2 to the other half), share the PED and revenue results 3, Task 3 9, Task 4 7, Task 5 9, exit ticket 5. Task 3 needs results for both products, so share them before it starts; ask for about 60 words in Task 5 or set it as homework. This timing has not been trialled. The 45 minutes are task time only, so in a real 45-minute period, with handing out and feedback, move the exit ticket to homework.

Support and extension

Support: show the percentage change method on the board (change / original x 100) and complete Task 1a with the class. Extension: ask students to predict, without calculating, whether a ticket price rise from £10 to £11 would raise or lower revenue if PED stays at -1.5, then check it with the exit ticket.

Answers: starter and Task 1

Indicative guidance for teachers. This is not an exam board mark scheme and carries no marks or grades.

Starter: (i) elastic; (ii) inelastic; (iii) unit elastic; (iv) elastic. Unit elastic note: at a PED of exactly -1 a small price change leaves revenue broadly unchanged.

Task 1	Answer	Working
a. Percentage change in price	-10.0%	(9 - 10) / 10 x 100
a. Percentage change in quantity	+15.0%	(2,300 - 2,000) / 2,000 x 100
b. PED	-1.5; elastic	+15.0 / -10.0; the size is above 1
c. Ticket revenue	£20,000 before; £20,700 after; +£700	10 x 2,000; 9 x 2,300

Answers: Task 2

Task 2	Answer	Working
a. PED	-0.4; inelastic	Price +10.0% (0.60 / 6.00); quantity -4.0% (-40 / 1,000); -4.0 / +10.0
b. Combo revenue	£6,000 before; £6,336 after; +£336	6.00 x 1,000; 6.60 x 960

c. When demand is inelastic, the percentage fall in quantity is smaller than the percentage rise in price, so total revenue rises. When demand is elastic, the percentage fall in quantity is larger, so revenue falls. These results are consistent with the rule: a price cut raised revenue where demand was elastic (Task 1) and a price rise raised revenue where demand was inelastic (Task 2). Accept an answer that uses the numbers.

Answers: Task 3

£ per week	Before	After	Change
Ticket revenue	20,000	20,700	+700
Ticket variable costs	3,000	3,450	+450
Ticket contribution	17,000	17,250	+250
Combo revenue	6,000	6,336	+336
Combo variable costs	2,500	2,400	-100
Combo contribution	3,500	3,936	+436

Working: ticket variable costs 1.50 x 2,000 = 3,000 and 1.50 x 2,300 = 3,450; combo variable costs 2.50 x 1,000 = 2,500 and 2.50 x 960 = 2,400. Task 3b: ticket revenue rises by £700 but contribution rises by only £250, because the 300 extra tickets each carry £1.50 of variable cost (300 x 1.50 = £450). Revenue ignores the extra costs of selling more (this assumes fixed costs do not change). For the combo the pattern reverses: contribution rises by more than revenue (+£436 against +£336) because 40 fewer combos save £100 of variable cost, so a price rise on inelastic demand can raise profit by more than revenue.

Answers: Task 4 and Task 5

Indicative guidance, not an exam board mark scheme.

Task 4 indicative answer. Two strong factors, each tied to Starlight. (1) Availability of substitutes: a customer can replace a cinema ticket by streaming a film at home or going to another cinema, but inside the foyer the combo may have fewer immediate substitutes, so demand could be less elastic. (2) Proportion of income: the combo takes a tiny share of a customer's budget, so a 60p rise may be less noticeable, whereas a family buying several tickets spends a larger share. Other valid factors influencing PED are acceptable if applied to the case.

Task 5: indicative decision guidance

A well supported answer judges the manager's claim that both changes raise revenue and contribution against the figures. Supportive points: the ticket cut raised revenue by £700 with PED of -1.5, and the combo rise raised revenue by £336 with PED of -0.4, both consistent with the PED and revenue rule. Contribution also rose in both tests (by £250 and £436), though the ticket gain is much smaller than the revenue gain suggests. Limits: each PED comes from a single one-week test; the cut for tickets is only 10% so the estimate may not hold for bigger cuts; demand may differ by film, day or season; customers who pay less for tickets may buy more snacks, so the two products are linked; fixed costs, capacity (are 2,300 seats available?) and competitors' reactions are not in the data; a lower ticket price may affect brand image. The tests are idealised: a one-week test where nothing else changes is rare, and a ticket PED of -1.5 is a teaching figure, not market evidence. A conclusion such as "the evidence supports the plan but should be tested for longer" is strong.

Look for correct figures, a clear link between PED and revenue, a reason why each PED estimate may be unreliable and a conclusion tied to the evidence. Do not award a grade or percentage from these tasks.

Answers: exit ticket

a. Price +10.0% (1 / 10); quantity -15.0% (-300 / 2,000); PED = -15.0 / +10.0 = -1.5 (elastic).

b. Revenue falls from £20,000 (10 x 2,000) to £18,700 (11 x 1,700), a fall of £1,300. Because demand is elastic, the percentage fall in quantity is larger than the percentage rise in price.

Sources and limits

Curriculum source: Pearson Edexcel Level 3 Advanced GCE in Business specification, Issue 1, September 2014, Theme 1, section 1.2.4 (printed page 7) and 2.2.3(a) (printed page 15). The specification verifies the topic scope only. The tasks, figures, timings and guidance are original TBS teaching design.